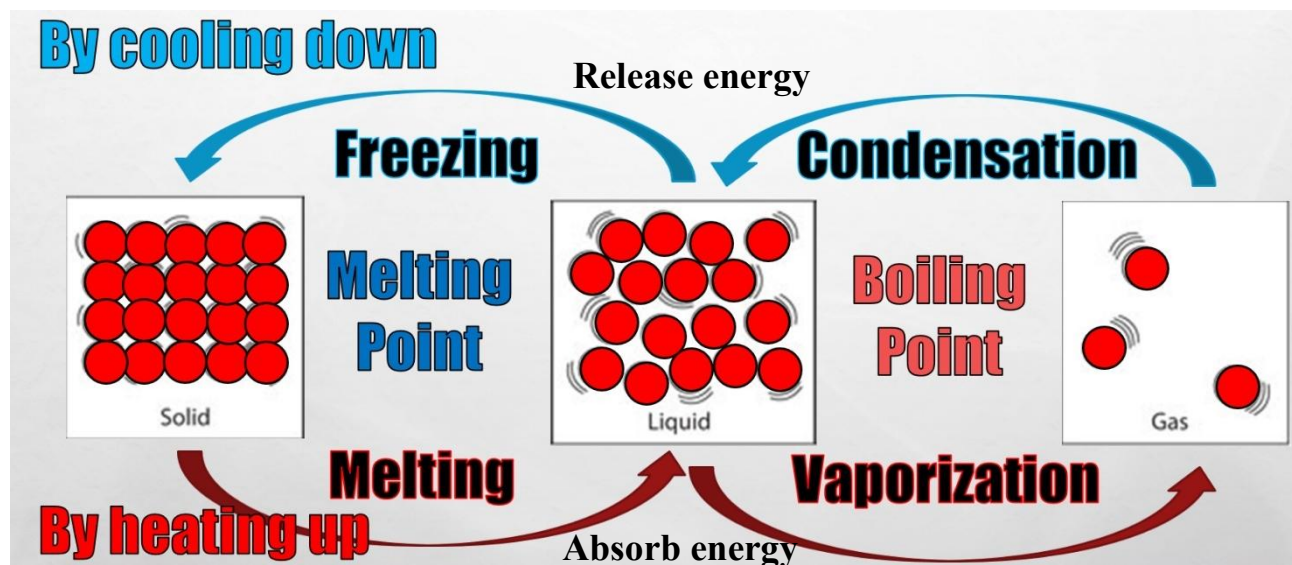




Specific latent heat of fusion of ice

The specific latent heat of fusion of ice is $334\,000\text{ J kg}^{-1}$ = energy needed (334 000 J) to change 1 kg ice at 0°C to 1 kg water at 0°C .

$$E = ml_f$$

Specific latent heat of vaporization of water

The specific latent heat of vaporization of water is $2\,260\,000\text{ J kg}^{-1}$ = energy needed (2 260 000 J) to change 1 kg water at 100°C to 1 kg steam at 100°C .

$$E = ml_v$$

- The **energy** absorbed or released by a substance during a change of state is called **latent heat**.

(i) Fusion/melting:

- A substance changes from a **solid state** to a **liquid state** at the melting point.
- A substance **absorbs** energy but its temperature does not change.

(ii) Vaporization:

- A substance **absorbs** energy and changes from a **liquid state** to a **gaseous state**.
- Boiling** and **evaporation** are the two types of vaporization. Boiling **only** occurs **at the boiling point** but evaporation can occur **below the boiling point**.
- When a substance boils, its temperature does not change.

(iii) Freezing (or solidification):

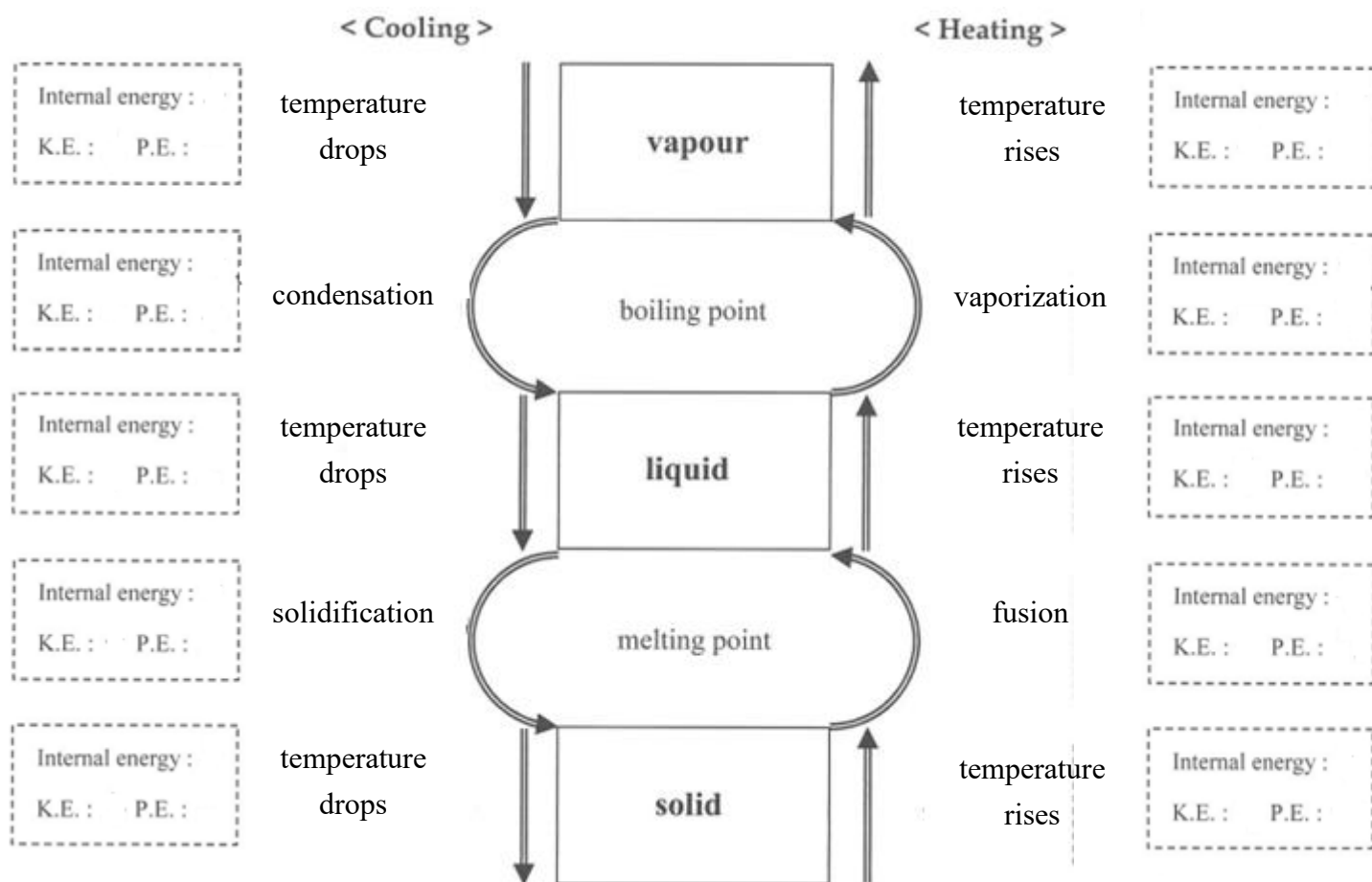
- A substance changes from a **liquid state** to a **solid state** at the melting point
- A substance **release** energy but its temperature does not change

(iv) Condensation:

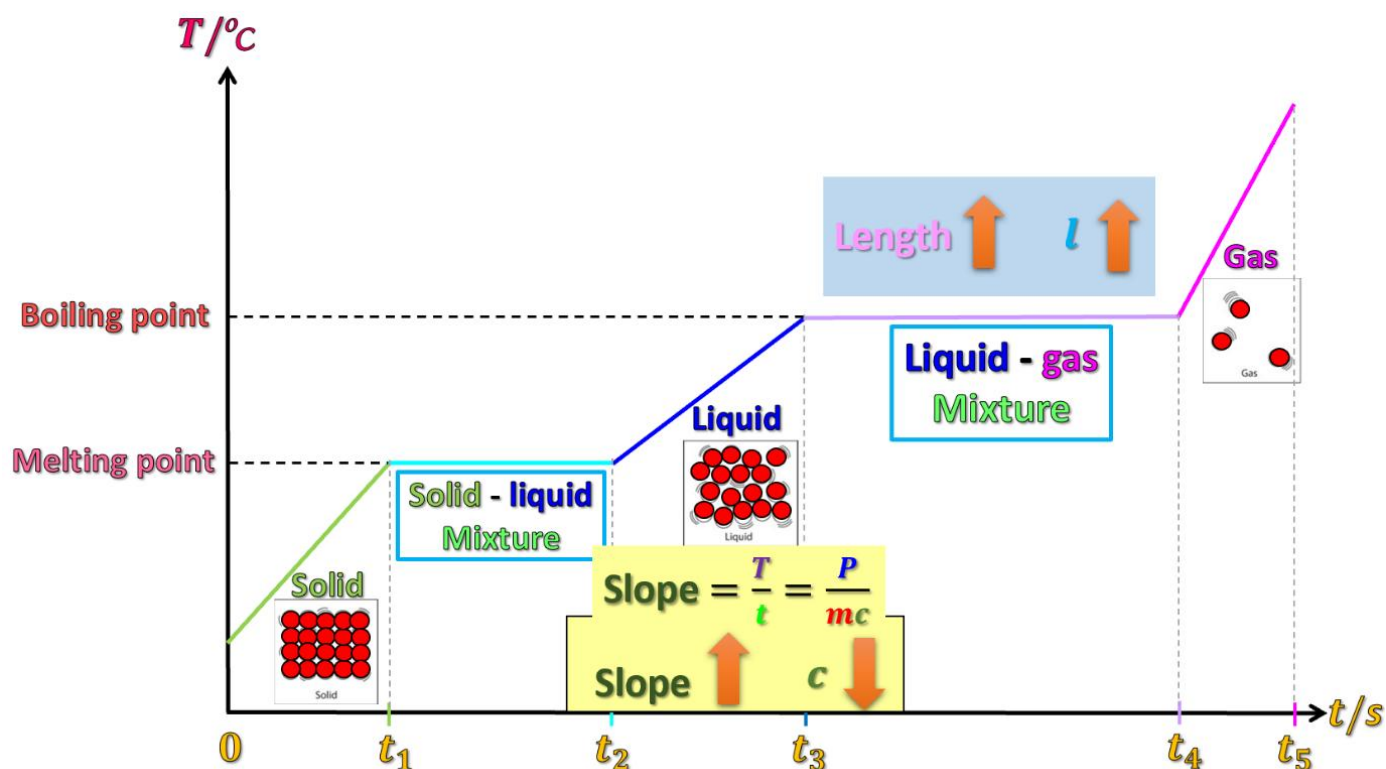
- A substance changes from a **gaseous state** to a **liquid state** and **release** energy

- Molecular potential energy arises due to the attractive forces between molecules.
- When a substance changes from solid to liquid or liquid to gas, the molecular **potential energy increase**.
- The internal energy of a body is the **sum** of the **kinetic energy** and **potential energy** of all its molecules. It is the total energy stored in the body.
- Internal energy depends on:
 - (1) temperature of the substance (average molecular KE)
 - (2) State of the substance (average molecular PE)
 - (3) Mass of the substance (no. of molecules)

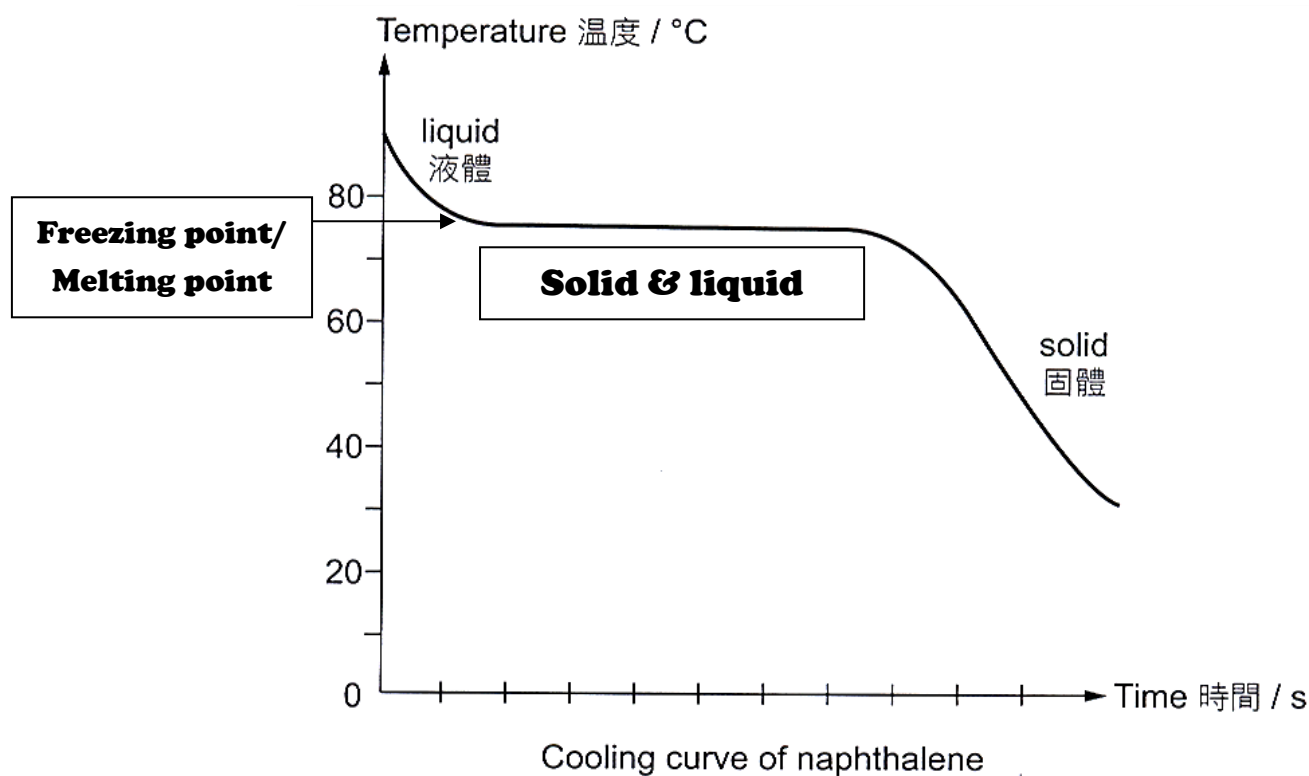
Summary of energy changes



Heating curve



Cooling curve



Evaporation and boiling

	Evaporation	Boiling
Differences	Occurs at any temperature below the boiling point	Occurs at the boiling point
	Occurs on the liquid surface	Occurs throughout the liquid
	No bubbles are formed.	Bubbles are formed inside the liquid
	Energy is taken from the liquid itself or its surroundings	Energy is taken from the heat source
Similarity	Latent heat of vaporization is absorbed in both processes	

Molecular motion during evaporation:

- (i) In a liquid, the molecules keep bumping into each other. Some molecules gain KE while other lose KE.
- (ii) When the molecules at the surface of a liquid gain enough kinetic energy, they may **escape** and become vapour molecules.
- (iii) The **molecules remaining** in the liquid have **lower average kinetic energy**. Therefore, the **temperature** of the liquid **drops**.

Remarks: There are several factors that **increase** the rate of evaporation:

(1) temperature increase

As liquid temperature increase, molecules move faster on average so that more molecules can escape.

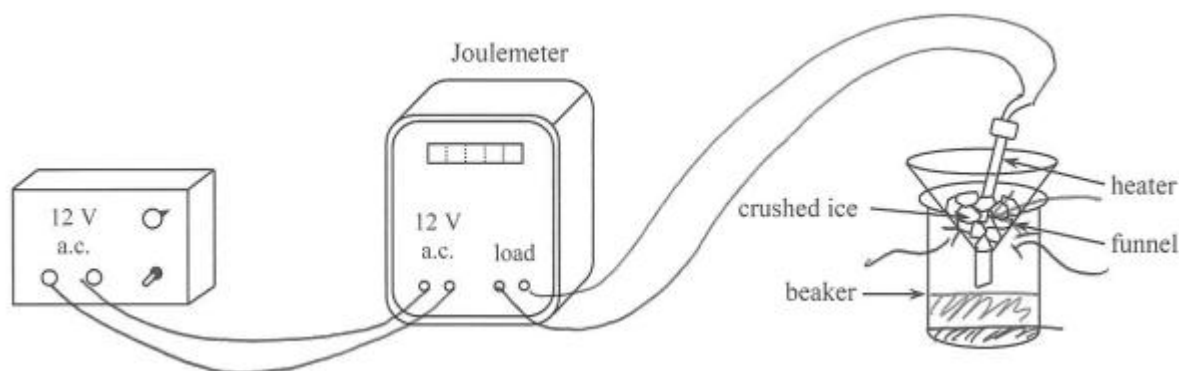
(2) Surface increase

The larger the surface area, the more molecules at the liquid surface and can escape.

(3) Density of vapour decrease

As vapour density in the environment decrease, vapour molecules have lower chance of returning to liquid

Measurement of specific latent heat of fusion of ice



(i) Procedure of the experiment:

- Put some crushed ice into a funnel placed above a beaker.
- A heater connected to a power supply through a joulemeter is put into the crushed ice. The energy supply by the heater E is recorded by the joulemeter.
- Measure the mass of water m inside the beaker below the funnel.
- The specific latent heat of ice is then found by $l_f = \frac{E}{m}$

(ii) Precautions of the experiment:

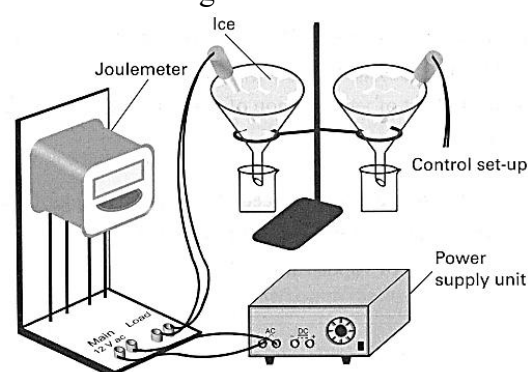
- The ice used should be **crushed** to increase the contact surface area of the ice and ensure good thermal contact between the ice and the heater.
- The ice should be **melting** to ensure no energy is used to raise the temperature of the ice before melting.
- The heater should be **completely immersed** into crushed ice.

(iii) Deviation of the result:

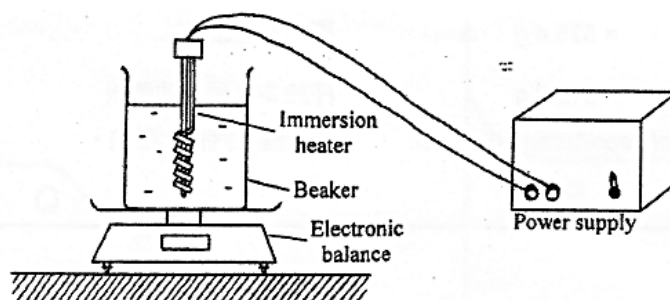
- The ice used is not at its melting point but at a lower temperature
 $\therefore$ extra energy is needed to raise the temperature of ice to its melting point.
 $\therefore$ the measured value would be **greater than** the actual value.
- **Heat gained** from the surroundings
 $\therefore$ mass of ice melted would be greater
 $\therefore$ the measured value would be **smaller than** the actual value.

(iv) Improvement of the experiment by using control set up:

- To find out the amount of ice melted due to **heat gained** from the surroundings
- To find out the amount of ice actually melted by the heater.



Measurement of specific latent heat of vaporization of water



(i) Procedure of the experiment:

- A beaker containing water is placed on the electronic balance.
- A heater of known power P is immersed into the water.
- When the water boils, a reading of the balance is recorded.
- After a certain period of time t recorded by stop watch, the reading of balance is taken again so that the mass of water m can be found.
- The energy E supplied by the heater is found by $E = Pt$
- The specific latent heat of vaporization of water is then found by $l_v = \frac{E}{m}$

(ii) Precautions of the experiment:

- The heater should be completely immersed into the water
- The beaker should be wrapped by insulating material to reduce heat lost to the surroundings

(iii) Deviation of the result:

- Some energy is lost to the surroundings
 $\therefore$ extra energy is supplied by the heater.
 $\therefore$ the measured value would be **greater than** the actual value.
- Steam condenses at upper part of the apparatus and drips back to the beaker
 $\therefore$ More energy is supplied by the heater
 $\therefore$ the measured value would be **greater than** the actual value.
- Boiling may be too vigorous that some water jump out from beaker.
 $\therefore$ More water would leave the beaker
 $\therefore$ The measured value would be **smaller than** the actual value.

1. How much energy is required to change 0.5 kg of water at 100°C to steam at 100°C ? (specific latent heat of vaporization of water = $2\,260\,000\text{ J kg}^{-1}$)

$$\begin{aligned} E &= ml \\ &= (0.5)(2260000) \\ &= 1\,130\,000\text{ J} \end{aligned}$$

2. If $2 \times 10^5\text{ J}$ energy is needed to melt 4 kg of a solid at its melting point 70°C for 10 minutes. Find the specific latent heat of fusion.

$$\begin{aligned} E &= ml \\ 2 \times 10^5 &= (4)(l) \\ l &= 50\,000\text{ J kg}^{-1} \end{aligned}$$

3. A $1\,000\text{ W}$ electric heater is used to vaporize water. How much water, at 100°C , can be changed to steam in 3 minutes by the heater? You are given that the latent heat of vaporization of water is $2.26 \times 10^6\text{ J kg}^{-1}$.

The energy supplied by the heater = The energy gained by the water

$$P \times t = l_v m$$

$P = 1000\text{ W}$, $t = 3\text{ min} = 3 \times 60\text{ s} = 180\text{ s}$, $l_v = 2.26 \times 10^6\text{ J kg}^{-1}$, $m = ?$

$$(1000\text{ W})(180\text{ s}) = (2.26 \times 10^6\text{ J kg}^{-1})(m)$$

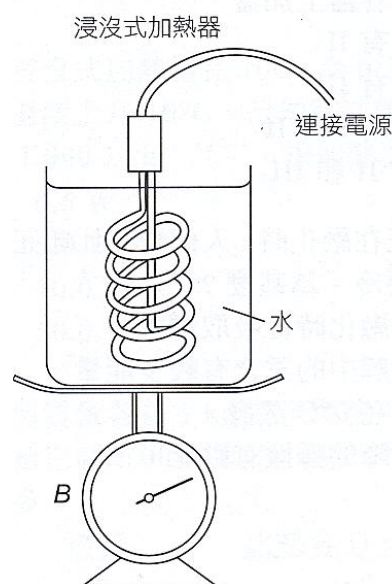
$$m = 0.08\text{ kg}$$

The mass of water vaporized is 0.08 kg.

4. An immersion heater of power 500 W is used to boil 1.1 kg of water. After boiled 10 minutes, the readings of the balance decrease from 1.1 kg to 1 kg.

- (a) How much energy the heater has given out in 10 minutes?
(b) From the experiment, calculate the specific latent heat of vaporization.

$$\begin{aligned} \text{(a)} \quad E &= Pt \\ &= (500)(10 \times 60) \\ &= 300\,000\text{ J} \\ \text{(b)} \quad E &= ml \\ 300\,000 &= (1.1 - 1)(l) \\ &= 3\,000\,000\text{ J kg}^{-1} \end{aligned}$$



5. A 556.7 W heater is used to melt 10 kg ice.
(a) After 1 hour, find the mass of ice remained.
(b) After 1 hour, Estimate the final temperature.
(c) How long does it take to melt all the ice?
(d) After 2 hours, calculate the temperature.
(specific latent heat of fusion of ice = $334\,000\text{ J kg}^{-1}$)
(specific heat capacity of water = $4\,200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$)

$$\begin{aligned} Pt &= ml \\ 556.7 (60 \times 60) &= (m) (3.34 \times 10^5) \\ m &= 6\text{ kg} \\ \therefore \text{the mass of ice remained is } &4\text{ kg.} \end{aligned}$$

$$0\text{ }^{\circ}\text{C}$$

$$\begin{aligned} Pt &= ml \\ 556.7 (t) &= (10) (3.34 \times 10^5) \\ t &= 5999.641\text{ s} \end{aligned}$$

$$\begin{aligned} Pt &= ml + mc \Delta T \\ 556.7 (2 \times 60 \times 60) &= (10) (3.34 \times 10^5) + (10) (4200) \Delta T \\ \Delta T &= 15.9\text{ }^{\circ}\text{C} \end{aligned}$$

6. A 1 000 W heater is used to heat 5 kg ice. After 1 hour, Calculate the temperature of the water.
(specific latent heat of fusion of ice = $334\,000\text{ J kg}^{-1}$)
(specific heat capacity of water = $4\,200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$)

$$\begin{aligned} Pt &= ml + mc \Delta T \\ 1000 (1 \times 60 \times 60) &= (5) (334\,000) + (5) (4200) \Delta T \\ \Delta T &= 91.9\text{ }^{\circ}\text{C} \end{aligned}$$

7. How much energy is needed to change $0\text{ }^{\circ}\text{C}$, 0.5 kg water to $100\text{ }^{\circ}\text{C}$ steam?
(specific latent heat of vaporization of water = $2.26 \times 10^6\text{ J kg}^{-1}$)
(specific heat capacity of water = $4\,200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$)

$$\begin{aligned} E &= ml + mc \Delta T \\ E &= (0.5) (2.26 \times 10^6) + (0.5) (4200) (100) \\ E &= 1\,340\,000\text{ J} \end{aligned}$$

8. How much energy is given up when 25 g of steam, at 100°C , is condensed and cooled to 30°C ?
(specific latent heat of vaporization of water = $2.26 \times 10^6 \text{ J kg}^{-1}$)
(specific heat capacity of water = $4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$)

The total energy released = The latent heat of vaporization + The energy released from the water

$$= l_v m + cm\Delta T$$

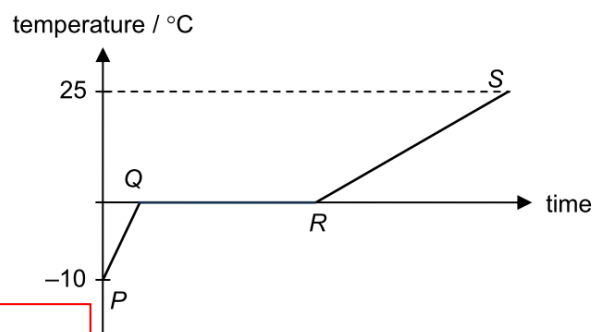
$$c = 4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}, l_v = 2.26 \times 10^6 \text{ J kg}^{-1}, m = 25 \text{ g} = 0.025 \text{ kg}, \Delta T = (100^{\circ}\text{C} - 30^{\circ}\text{C}) = 70^{\circ}\text{C}$$

$$= (2.26 \times 10^6 \text{ J kg}^{-1})(0.025 \text{ kg}) + (4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1})(0.025 \text{ kg})(70^{\circ}\text{C})$$

$$= 63850 \text{ J}$$
The amount of energy given out is 63850 J.

9. The following graph shows how the temperature of 0.5 kg of ice changes as it is heated. What is the total energy absorbed by the ice from P to S?

Given: specific heat capacity of water = $4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$
specific heat capacity of ice = $2050 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$
specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$



$$E = mc\Delta T + ml + mc\Delta T$$

$$E = 0.5 \times 2050 \times [0 - (-10)] + 0.5 \times 3.34 \times 10^5 + 0.5 \times 4200 \times (25 - 0)$$

$$= 1.025 \times 10^4 + 1.67 \times 10^5 + 5.25 \times 10^4$$

$$= 2.30 \times 10^5 \text{ J}$$

10. The specific latent heat of fusion of the above substance is 50000 J kg^{-1} . Assume no heat loss to surrounding find the specific heat capacity of the left substance.

$$Pt = mc\Delta T \text{ and } Pt = ml$$

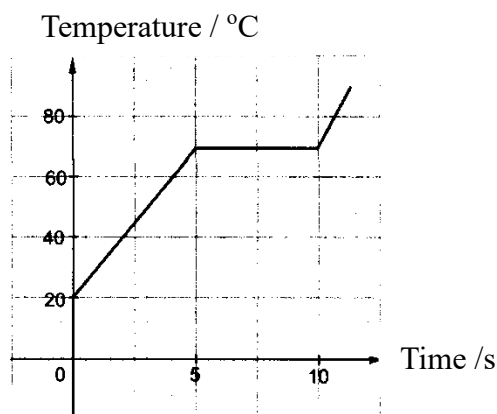
$$\therefore \text{same } P \text{ and } t$$

$$\therefore mc\Delta T = ml$$

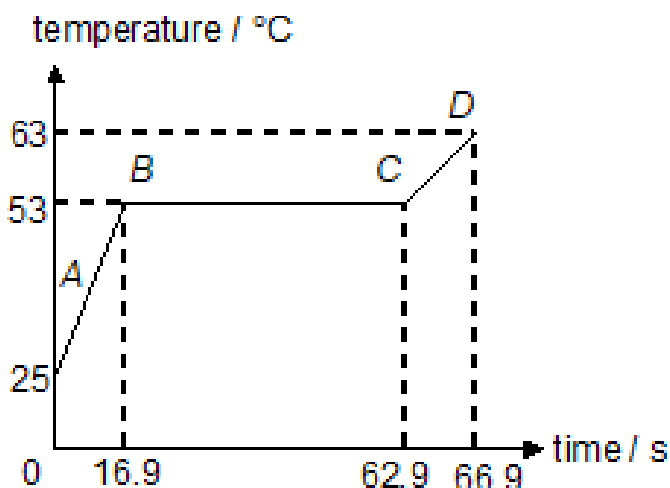
$$l = c\Delta T$$

$$50000 = c(70 - 20)$$

$$c = 1000 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$$



11. In an experiment, 0.5 kg of substance X is heated by a 2000 W heater. The graph below shows the temperature variation of the substance when changing from solid state to liquid state.
- (a) What would be the state(s) of the substance in the following region of the graph?
- BC
 - CD
- (2 marks)
- (b) Calculate the specific heat capacity of substance X in solid state.
- (2 marks)
- (c) Calculate the specific latent heat of fusion of substance X .
- (2 marks)
- (d) Suppose the experiment is repeated with the power of the heater halved. Sketch, on the same graph, how the new temperature of substance X varies with time.
- (4 marks)



- (a) (i) The substance exists in both solid and liquid states. (1A)*
(ii) The substance exists in liquid state. (1A)*

- (b) Applying $Q = mc\Delta T$, the specific heat capacity of substance X in solid state is given by ↵

$$(2000)(16.9) = (0.5)(c)(53 - 25)$$

$$c = 2414 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$$

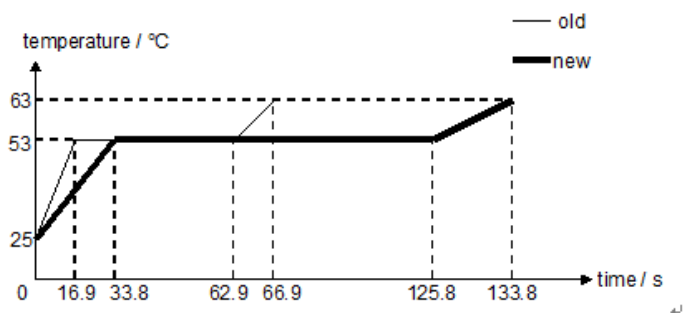
(1M+1A)*

- (c) Applying $Q = mL_f$, the specific latent heat of fusion of substance X is given by ↵

$$(2000)(62.9 - 16.9) = (0.5)(L_f)$$

$$L_f = 184\,000 \text{ J kg}^{-1}$$

- (d) If the power is halved, the time spent in each section of the curve will be doubled. ↵



(1A for correct melting point + 1A for the same initial temperature + 2A for correct slopes)*

12. A boy puts 0.5 kg of water at 25 °C into the freezer compartment of a refrigerator. The water freezes and the temperature of the ice drops to -8 °C.

(a) Sketch a graph to show how the temperature of the water changes with time.

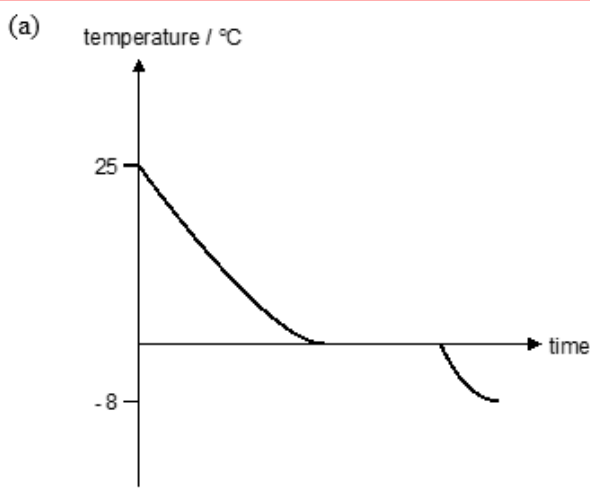
(3 marks)

(b) Find the total energy removed from the water and the ice.

(The specific heat capacities of water and ice are 4200 J kg⁻¹ °C⁻¹ and 2100 J kg⁻¹ °C⁻¹ respectively.

The specific latent heat of fusion of water is 3.34 × 10⁵ J kg⁻¹.)

(4 marks)



(1A for the initial and final temperatures + 1A for the shape of the graph + 1A for the labels of the axes)

(b) Applying $Q = mc\Delta T$, the energy removed in cooling down the water from 25 °C to 0 °C is

$$Q = 0.5 \times 4200 \times (25 - 0)$$

$$= 5.25 \times 10^4 \text{ J}$$

(1M)

Applying $Q = mL_f$, the energy removed when the water at 0 °C changes to ice at 0 °C is

$$Q = 0.5 \times 3.34 \times 10^5$$

$$= 1.67 \times 10^5 \text{ J}$$

(1M)

Applying $Q = mc\Delta T$, the energy removed in cooling down the ice from 0 °C to -8 °C is

$$Q = 0.5 \times 2100 \times [0 - (-8)]$$

$$= 8400 \text{ J}$$

(1M)

Hence, the total energy removed is

$$(5.25 \times 10^4 + 1.67 \times 10^5 + 8400) \text{ J} = 2.279 \times 10^5 \text{ J}$$

(1A)

13. A tank containing 6 kg of water at 30°C is poured on a large block of ice at 0°C .

If the final temperature of the water is 0°C , how much ice melts?

(specific latent heat of fusion of ice = $3.33 \times 10^5 \text{ J kg}^{-1}$)

(specific heat capacity of water = $4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$)

The energy lost by the water = The energy gained by the ice

$$m_w c_w \Delta T = l_f m_i$$

$$m_w = 6 \text{ kg}, c_w = 4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}, l_f = 3.33 \times 10^5 \text{ J kg}^{-1}, \Delta T = (30^{\circ}\text{C} - 0^{\circ}\text{C}) = 30^{\circ}\text{C}, m_i = ?$$

$$(6 \text{ kg})(4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1})(30^{\circ}\text{C}) = (3.33 \times 10^5 \text{ J kg}^{-1})(m_i)$$

$$m_i = 2.27 \text{ kg}$$

The mass of ice melted is 2.27 kg.

14. A 20 g of ice cube, at 0°C is dropped into 300 g of water at 25°C .

What is the final temperature of the water?

(specific latent heat of fusion of ice = $3.33 \times 10^5 \text{ J kg}^{-1}$)

(specific heat capacity of water = $4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$)

Let the final temperature of water be $T^{\circ}\text{C}$.

The energy lost by the water = The energy gained by the ice

= the latent heat required to convert the ice at 0°C to water 0°C +

the energy required to raise the temperature of the water from 0°C to T

$$m_w c_w \Delta T_1 = l_f m_i + m_i c_w \Delta T_2$$

$$m_w = 300 \text{ g} = 0.3 \text{ kg}, m_i = 20 \text{ g} = 0.02 \text{ kg}, c_w = 4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}, l_f = 3.33 \times 10^5 \text{ J kg}^{-1},$$

$$\Delta T_1 = (25 - T), \Delta T_2 = (T - 0)$$

$$(0.3)(4200)(25 - T) = (3.33 \times 10^5)(0.02) + (0.02)(4200)(T - 0)$$

$$31500 - 1260T = 6660 + 84T$$

$$24840 = 1344T$$

$$T = 18.5^{\circ}\text{C}$$

The final temperature of the is 18.5°C

15. A 100 g ice cube at 0°C is dropped into 800 g of Coke at 25°C . What is the final temperature of the Coke?

Given: The specific heat capacity of Coke = $4800 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$

The specific heat capacity of water = $4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$

The specific heat capacity of fusion of ice = $3.33 \times 10^5 \text{ J kg}^{-1}$

Let the final temperature of the Coke be $T^{\circ}\text{C}$.

The energy lost by the Coke = The energy gained by the ice

$$m_c c_c \Delta T_c = m_w l_f + m_w c_w \Delta T_w$$

$$m_c = 800 \text{ g} = 0.8 \text{ kg}, c_c = 4800 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}, \Delta T_c = (25^{\circ}\text{C} - T),$$

$$l_f = 3.33 \times 10^5 \text{ J kg}^{-1}, m_w = 100 \text{ g} = 0.1 \text{ kg}, c_w = 4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}, \Delta T_w = (T - 0) = T,$$

$$(0.8 \text{ kg})(4800 \text{ J kg}^{-1}^{\circ}\text{C}^{-1})(25^{\circ}\text{C} - T) = (0.1 \text{ kg})(3.33 \times 10^5 \text{ J kg}^{-1}) + (0.1 \text{ kg})(4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1})(T)$$

$$96000 - 3840T = (33300) + 420T$$

$$62700 = 4260T$$

$$T = 14.7^{\circ}\text{C}$$

The final temperature of the Coke is 14.7°C .

16. 1 kg of ice and 1 kg of steam are mixed together in an insulated chamber. The ice is at 0 °C while the steam is at 100 °C. Find the final temperature of the mixture. In which state is the mixture?

(Specific heat capacity of water = 4200 J kg⁻¹ °C⁻¹)

Specific latent heat of fusion of ice = 3.34 × 10⁵ J kg⁻¹

Specific latent heat of vaporization of water = 2.26 × 10⁶ J kg⁻¹)

Energy required to heat the ice from 0 °C to water at 100 °C

$$= 1 \times (3.34 \times 10^5) + 1 \times 4200 \times (100 - 0)$$

$$= 754\,000 \text{ J}$$

Energy released when all steam condenses

$$= 1 \times (2.26 \times 10^6)$$

$$= 2\,260\,000 \text{ J} > 754\,000 \text{ J}$$

∴ Only part of the steam condenses when all the ice is melted and heated to 100 °C.

∴ The mixture is in both liquid and gas states and is at 100 °C.

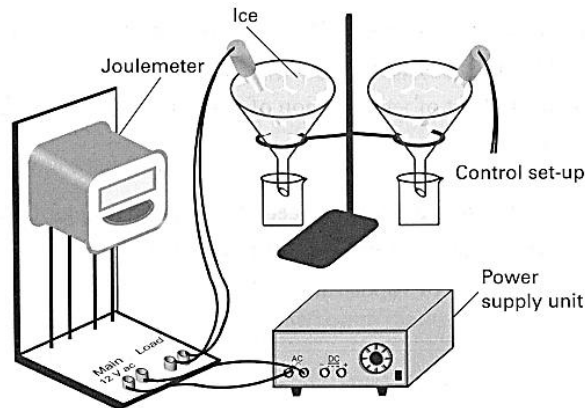
17. The following set-up is used to measure the specific latent heat of fusion of ice.

Mass of beaker A + Water = 0.14 kg

Mass of beaker B + Water = 0.09 kg

Initial joulemeter reading = 23 876

Final joulemeter reading = 40 896



- (a) Using the data, find the specific latent heat of fusion of ice.
(b) Explain why the control experiment is necessary for this experiment.

(a) The energy supplied by the heater (Q) = 40 896 – 23 876 = 17 020 J

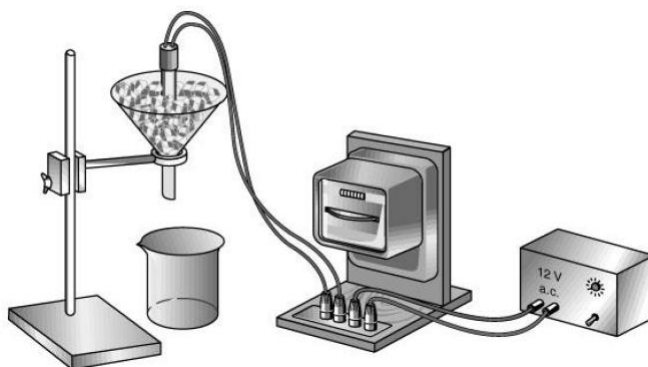
The mass of water melted by the heater (m) = 0.14 kg – 0.09 kg = 0.05 kg

The specific latent heat of fusion of ice (l_f) = $\frac{Q}{m} = \frac{17\,020 \text{ J}}{0.05 \text{ kg}} \text{ J kg}^{-1} = 3.4 \times 10^5 \text{ J kg}^{-1}$

The specific latent heat of fusion of ice is $3.4 \times 10^5 \text{ J kg}^{-1}$.

(b) We can find out how much ice is melted from the energy absorbed from the surroundings.

18. Jerry performs an experiment to determine the specific latent heat of fusion of ice.



- How can Jerry ensure that the ice used is at 0 °C? (1 mark)
- Jerry obtains the following data in the experiment:
Initial joulemeter reading = 35 000 J
Final joulemeter reading = 46 400 J
Mass of water collected in beaker = 0.04 kg
Find the specific latent heat of fusion of ice. (2 marks)
- The result of the experiment is inaccurate because some ice was melted by absorbing energy from the surroundings. Describe how the amount of ice melted by the surroundings can be measured. (4 marks)
- Give three reasons why ice is good for cooling drinks. (3 marks)

- He should use melting ice.
- Energy supplied by heater = energy gained by ice
 $46\,400 - 35\,000 = 0.04 \times l_f$
 $l_f = 2.85 \times 10^5 \text{ J kg}^{-1}$
- To measure the amount of ice melted at room temperature, a set of control apparatus is needed.
The control apparatus is identical to the experimental apparatus except that the immersion heater is not turned on.
Before putting the beakers underneath the funnels and switching on the heater in the experimental apparatus, make sure that the drip rates of the two funnels are about the same.
After switching off the heater, do not remove the beakers immediately. Wait until the drip rates are about the same again.
- Ice can be prepared easily.
Ice has a large specific latent heat of fusion.
It melts at room temperature.
(Or other reasonable answers)

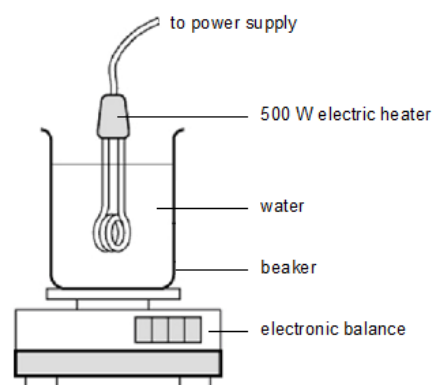
19. The set-up below is used to determine the specific latent heat of vaporization of water. When the water starts to boil, the balance reading is recorded. After 2 minutes, the balance reading is recorded again.

The following are the results obtained.

Initial balance reading = 652.1 g

Final balance reading = 627.3 g

- Find the specific latent heat of vaporization of water.
- The standard value of the specific latent heat of vaporization of water is $2.26 \times 10^6 \text{ J kg}^{-1}$. Suggest two reasons why the value in (a) is different from the standard value.
- John suggests that covering the beaker with a lid can improve the accuracy of the experiment. Do you agree? Explain briefly.



- Assume that there is no energy loss to the surroundings. By the law of conservation of energy
energy supplied by the heater = energy gained by the water

$$500 \times 2 \times 60 = \frac{652.1 - 627.3}{1000} \times l_v \quad (1M+1A)$$

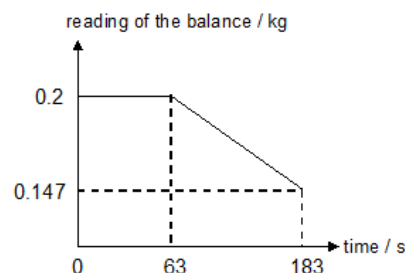
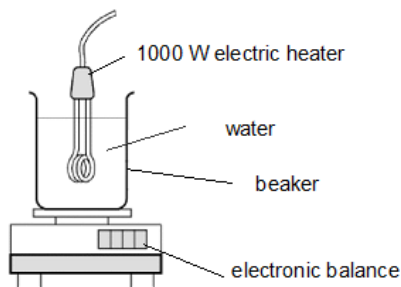
$$l_v \approx 2.42 \times 10^6 \text{ J kg}^{-1}$$

The specific latent heat of vaporization of water is $2.42 \times 10^6 \text{ J kg}^{-1}$.

- The experimental value is higher than the standard value. This is because some energy supplied by the heater is lost to the surroundings (1A). Besides, steam condensed on the side of the beaker may fall back into the water (1A).
- John is not correct (1A). If the beaker is covered by a lid, most of the water vapour will condense and fall back into the beaker (1A).

20. The set-up below is used to determine the specific latent heat of vaporization of water. The graph shows the variation of the reading of the electronic balance with time (the total mass of the empty beaker and the heater has already been eliminated). The initial temperature of the beaker of water is 25°C.

- How can you estimate the specific heat capacity of water from the graph? Briefly explain your answer and find the specific heat capacity of water.
- Find the specific latent heat of vaporization of water from the graph.
- Suggest an improvement in the set-up to increase the accuracy of the experiment.



- When the water temperature reaches 100°C, the water boils and the mass of water starts to decrease (1A). Thus, the rise in temperature, the energy absorbed and the mass of water are known when the water starts to boil (1A). We can estimate the specific heat capacity of water by applying equation

$$c = \frac{Q}{m\Delta T}, \text{ (1M)}$$

$$c = \frac{Q}{m\Delta T}$$

$$= \frac{(1000 \times 63)}{(0.2) \times (100 - 25)}$$

$$= 4200 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$$

(1A)

- Applying $Q = ml_v$

$$l_v = \frac{Q}{m}$$

$$= \frac{(1000)(183 - 63)}{0.2 - 0.147}$$

$$= 2.26 \times 10^6 \text{ J kg}^{-1}$$

(1M+1A)

Therefore, the specific latent of vaporization of water is $2.26 \times 10^6 \text{ J kg}^{-1}$.

- The accuracy can be improved by wrapping the beaker with cotton or using a foam cup instead of a beaker (1A).